

Academic Paper Analysis & Critique

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Chosen Cyber Technique: Man In The Middle - Evil Twin

Target Paper Citation: A. L. de S. Paula, E. K. Viegas, and A. O. Santin, "Passive Identification of Rogue Industrial Access Points Using a One-Class Machine Learning Model," in *Proceedings of the IEEE International Conference on Industrial Electronics Society (IECON)*, October 2024, pp. 1-6.

Paper Meta-Information & Research Group Focus

- Publication Venue: IEEE IECON 2024 (Annual Conference of the IEEE Industrial Electronics Society).
- Research Group & Affiliation: The Graduate Program in Computer Science at the Pontifical Catholic University of Parana, Brazil. The group is affiliated with research in the usage of ML in cybersecurity and especially in networks.
- Funding/Sponsorship: The research was partially sponsored by the Brazilian National Council for Scientific and Technological Development (CNPq), under grant numbers 304990/2021-3, 302937/2023-4, and 407879/2023-4.

Problem Statement & Literature gap analysis

- The Status Quo: The paper features 4 prior research papers that focus on using ML for detection of Evil Twin attacks, both using hardware signals and software characteristics.
- The Identified Limitations: the works have different limitations: some are too high level and cannot detect low level rogue APs, some are dependent of the Evil Twin being far from the real AP, and some need a lot of preparation (a lot of prior data or setup to the network) in order to function properly.
- The Core Research Question: The paper and the research try to create a detection system that is not dependent on the specific network, doesn't rely on mass data and can detect hardware based attacks as opposed to the rest of the researches.

Model Selection & Architectural Justification

Selected model / Architecture	Core technical mechanism	Reason for Selection
One-Class Support Vector Machine (OC-SVM)	Gets trained on "normal" data without getting	Most rogue AP detection models today rely on the

	exposed or classifying the abnormalities. Then, it can sense when a deviation from the normal data has occurred and detect anomalies.	premise that rogue APs behave in predicted and similar ways, but in reality many imposter APs have unique ways to detect them.
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- Baseline Comparisons: The paper doesn't feature the baseline models used for the comparison, if there were any.

Hyperparameters & Model Optimization

Hyperparameter Category	Specific Value/Setting	Technical Justification/ Selection Method
Kernel	Radial Basis Function	helps classify the behaviors into normal and abnormal. This specific function handles non-linear classification.
Regulization Parameter	Selected via cross-parameter grid search (explicit value is not given)	Controls how complex the decision of APs becomes, as opposed to being more lenient.
Gamma	Selected via cross-parameter grid search (explicit value is not given)	Helps to distinguish between false positives and actual rogue AP through the gamma value (the reach of each data entry).
Feature Scaling	Min-Max Normalization	Ensures all the features are given proper attention to.
Training Strategy	Focus only on legitimate APs	As mentioned above, this helps the model handle irregular data.

Data Pipeline & Implementation Workflow

- Raw data collection: The process begins with sniffing wireless packets over the network using Scapy.

- Feature Selection: The paper extracts 20 features - 9 of which are software based, and the other 11 are hardware based, which makes spoofing harder. The features that proved to be the most efficient for the training were Short GI 20Mhz, Cap MPDCU, data rates supported by the AP and Receive NDP.
- Data Normalization: The features are normalized using min-max normalization, meaning all were given the same importance and were set between 0 and 1. The extracted values are converted into a NumPy array.
- Model Processing: As shown above, the model chosen is a One-Class SVM, which gets trained received the normalized vectors and creates a “map” of all the legitimate APs, which helps it determine whether an AP is rogue or not.
- Classification Output: The OC-SVM outputs 1 for legitimate APs and -1 for rogue APs.

Experimental Environment & Quantitative Results

- Benchmark Dataset Used: The paper features and uses a new dataset named RRAD (Realistic Rogue AP Dataset), which contains 357 unique APs with roughly half rogue and half legitimate, and was split between training APs (~70%) and testing APs (~30%).
- Evaluation Metrics Defined: The paper shows the following metrics in a dedicated table: Sensitivity, Specificity, Precision, False Positive Rate, False Negative Rate, Accuracy, F1-Score and AUC.
- Reported Detection Performance: The paper doesn’t give a baseline model used, but the performance of the given model is:

Cyber Technique Model Variant	Accuracy	Precision	Recall (Sensitivity)	False Positive Rate
RRAD	0.8889	0.7949	1.00	0.1951

Novelty & Scientific Contribution

- The “Secret Sauce”: the approach used in the paper features a new perspective on rogue AP detection. The method chosen covers both hardware and software features, is adaptive to new attack techniques and is not focused on a specific attack pattern, it is fully passive and it doesn’t rely on lots of prior information. The paper also features a brand new dataset for detecting such rogue APs.
- Theoretical vs Practical Value: The model is fully usable for industrial networks, and is very strong at detecting rogue APs (it has a low false negative rate). Because the model is very cautious, the limitation it has is the high false positive rate, and apart

from that we don't have information about its performance in more popular datasets.

Actionable Engineering Takeaways for Our Project

- Evil Twin attacks can be detected using both software & hardware features, and many of the features tested in the paper's model can be used for our project.
- What to Adopt: checking both hardware & software features can make it easier to detect anomalies, training our model with the One Class classification can make our detection system be more versatile, and using Min-Max Normalization for our vectors can be beneficial.
- What to Avoid: First of all, we should test our detection model across multiple datasets and make sure it actually works thoroughly. We should also aim to avoid a high false positive rate like in the paper.
- Proposed Improvement: We would find which features are more relevant and harder to fake in Evil Twin attacks and make them more valued in the Min-Max normalization. We can also combine more than just OC-SVM, which can reduce the false-positive rate.